

PUNJAB FLOOD-TOLERANT SEEDS VARIETIES

Sector: Agriculture

Hazard addressed: Flood

Targeted assets: Sugarcane and Cotton



Description: This measure promotes the adoption of flood-tolerant and waterlogging-resistant seed varieties for sugarcane and cotton crops to reduce crop losses during flood seasons and excessive rainfall events. The intervention improves crop survival, stabilizes yields under flood-prone conditions, and enhances resilience of farming systems against climate-induced flooding and waterlogging.

Priority Districts: Bahawalpur, Khanewal, Lodhran, Multan, Muzaffargarh, Sahiwal, and Vehari.

TECHNICAL SPECIFICATIONS/DIMENSION



Unit of implementation: Hectare (ha) under flood-tolerant crop cultivation.

Scale assumptions: Applicable in flood-prone agricultural districts vulnerable to river flooding, excessive monsoon rainfall.

Key design parameters: Use of certified flood-tolerant sugarcane and cotton seed varieties; compatibility with local agro-climatic conditions; timely seed distribution; and climate-smart agronomic practices.

Time horizon: 2026 – 2032.

IMPACT



Risk reduction level: High.

Time to effectiveness: Immediate within the first crop season after adoption.

Expected impact: Improved survival and yield stability of sugarcane and cotton crops under flood and waterlogging conditions, reduced crop losses, and strengthened resilience of agricultural systems against flood hazards.

IMPLEMENTATION CONSIDERATIONS



Enabling conditions: Availability of certified seed supply chains, extension services, farmer awareness programs, and climate advisory systems.

Required institutions: Agriculture Department Punjab, Crop Research Institutes, Seed Certification Agencies, Agricultural Extension Services, and development partners.

Barriers: Weak seed distribution systems, affordability constraints, limited technical awareness, and resistance to adoption of improved seed varieties.

COST ASSUMPTIONS



Unit cost: Cotton: PKR 479 million (~ USD 1.71 million); Sugarcane: PKR 378 million (~USD 1.35 million).

Capital Expenditure (CAPEX): USD 0.9 million in the first year (2026), then USD 0.4 million per year (2027 – 2032). Includes procurement and distribution of certified flood-tolerant seeds, demonstration plots, extension services, and farmer awareness campaign.

Operational Expenditure (OPEX): N/A.

CO-BENEFITS & TRADE-OFFS



Environmental: Improved adaptation to flood-prone environments and reduced land degradation associated with prolonged waterlogging.

Social: Enhanced livelihood resilience and food security for farming communities vulnerable to flood impacts.

Economic: Reduced economic losses from flood-induced crop failures and improved long-term agricultural productivity.

Potential risks: Limited availability of certified flood-tolerant seeds, variable adoption rates, and performance uncertainty under extreme climatic conditions.

BENEFICIARIES



Primary beneficiaries: Sugarcane and cotton farmers, agricultural cooperatives, extension departments, and rural farming communities.

Distributional aspects: Particularly beneficial for flood-prone farming communities dependent on sugarcane and cotton cultivation.